



Physics-Informed Machine Learning for Cross-Surface Droplet Impact Prediction

A leave-one-surface-out study on laser-textured aluminium

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1 THE QUESTION

Can a model trained on 12 laser-textured surfaces predict how far a droplet spreads on a surface it has never seen?

Droplet-impact models are usually scored on random splits, where every test surface was also in training [2]. Surface design needs the opposite: a prediction **before** the surface is made.

1,498 impacts, 12 textured surfaces	125 impacts, smooth plate REF-H	1–160 mPa·s, water to 91 wt% glycerol
8–119 Weber number	9–4,435 Reynolds number	50–800 μm pitch; 6 or 25 μm deep

Public dataset of Može et al. [1], University of Ljubljana. Target: maximum spreading $\beta_{\max} = D_{\max}/D_0$.

KEY TAKEAWAYS

- ✓ Judge droplet models on **unseen surfaces**. Random splits hide the gap.
- Describe surfaces with quantities **every surface has**: φ makes the smooth plate the limit of the textured family.
- A **Gaussian process** beats 12 alternatives on unseen surfaces (0.040). Physics backbones and bigger models add no significant gain.
- Intervals hold on average, not per fluid, and fail on a different wettability. **Say so on the tool.**

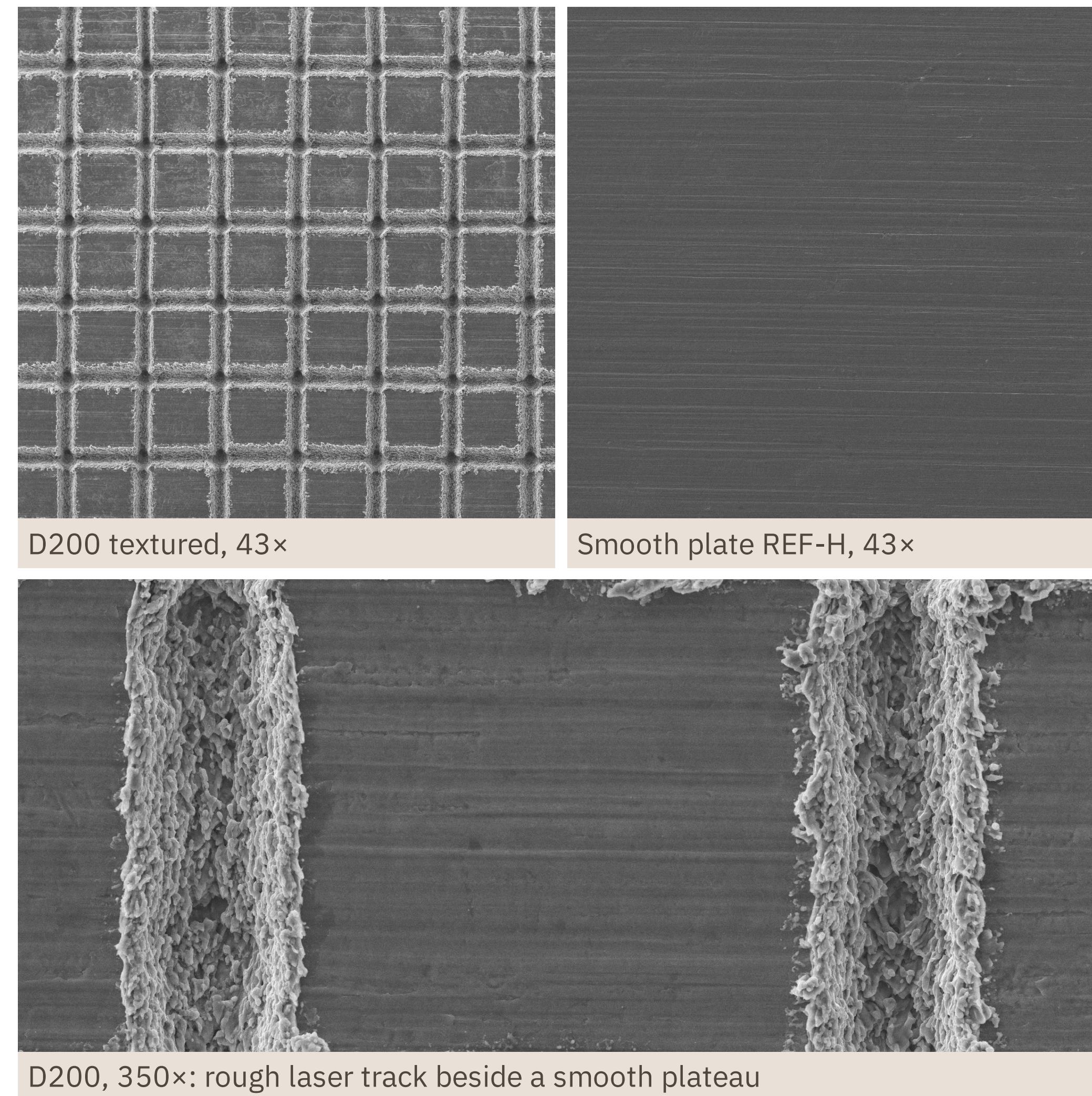
METHODS

2 Reading the surface from SEM

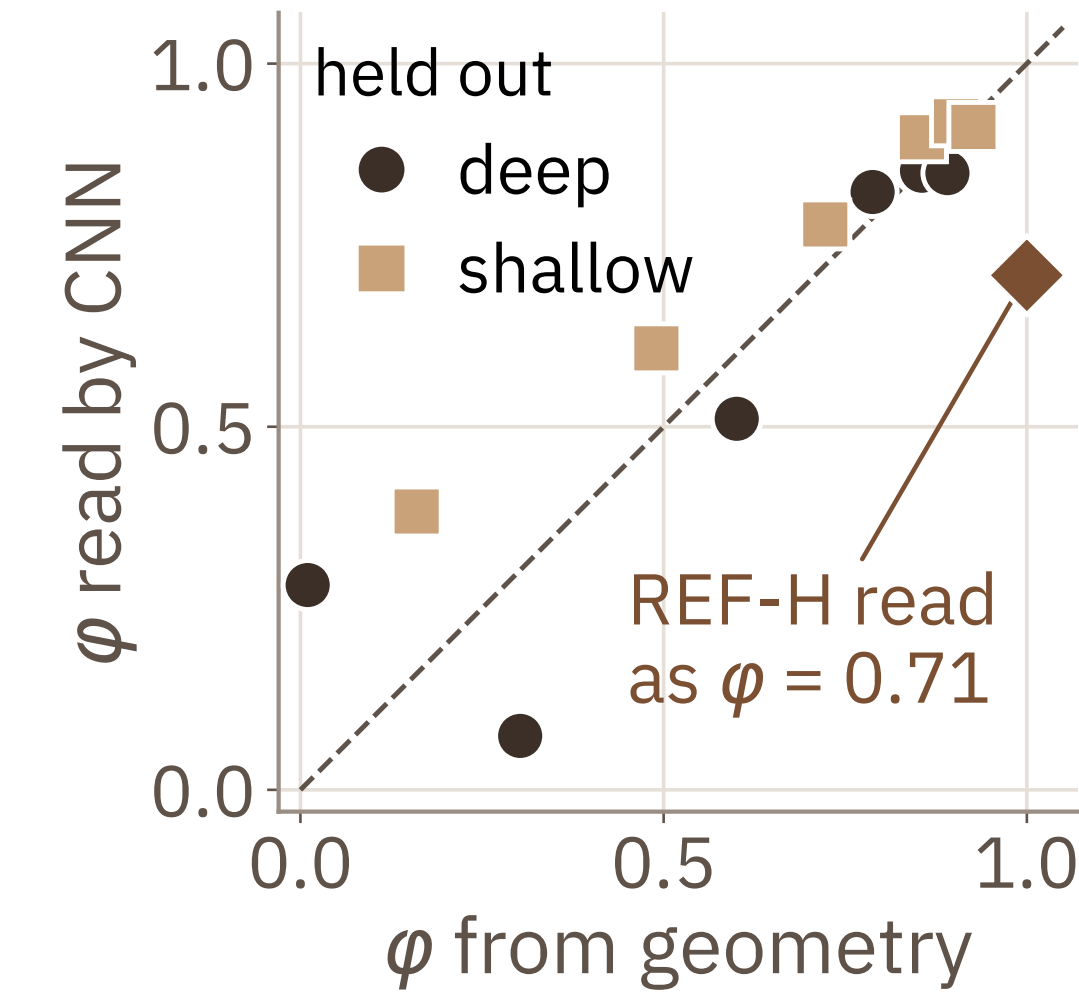
Plateaus between laser tracks look like the smooth plate. Track widths w from SEM (about 45 μm deep, 30 μm shallow) give every surface two descriptors:

$$\varphi = (1 - w/s)^2 \quad V_{\text{tex}} = (1 - \varphi) h$$

Textured φ runs from 0.01 (D50) to 0.93 (S800); **REF-H is simply $\varphi = 1$** . Invented spacing/depth codes moved REF-H error by up to 0.022; with φ , ± 10 μm in track width moves it by at most 0.005.



3 From SEM image to prediction



A 9,070-parameter CNN reads three SEM views (43×, 100×, 350×) of a surface **it never saw** and estimates φ and V_{tex} . The GP on these gives **0.038** vs 0.040, better on 8 of 12 surfaces but not significant. REF-H is read as $\varphi = 0.71$: no training surface is smoother than 0.93.

4 Physics inputs

$$Re = \frac{\rho V D_0}{\mu} \quad We = \frac{\rho V^2 D_0}{\sigma} \quad \beta_{\max} Re^{-1/5} = \frac{\sqrt{P}}{A + \sqrt{P}}$$

Recomputed Re and We match the dataset to 5×10^{-7} ; Oh adds nothing new. The Laan scaling law [3] ($P = We Re^{-2/5}$) was derived on wettable surfaces: here it reaches only **$R^2 = 0.71$** and misses unseen surfaces by **11%** on average.

5 Leakage-safe evaluation

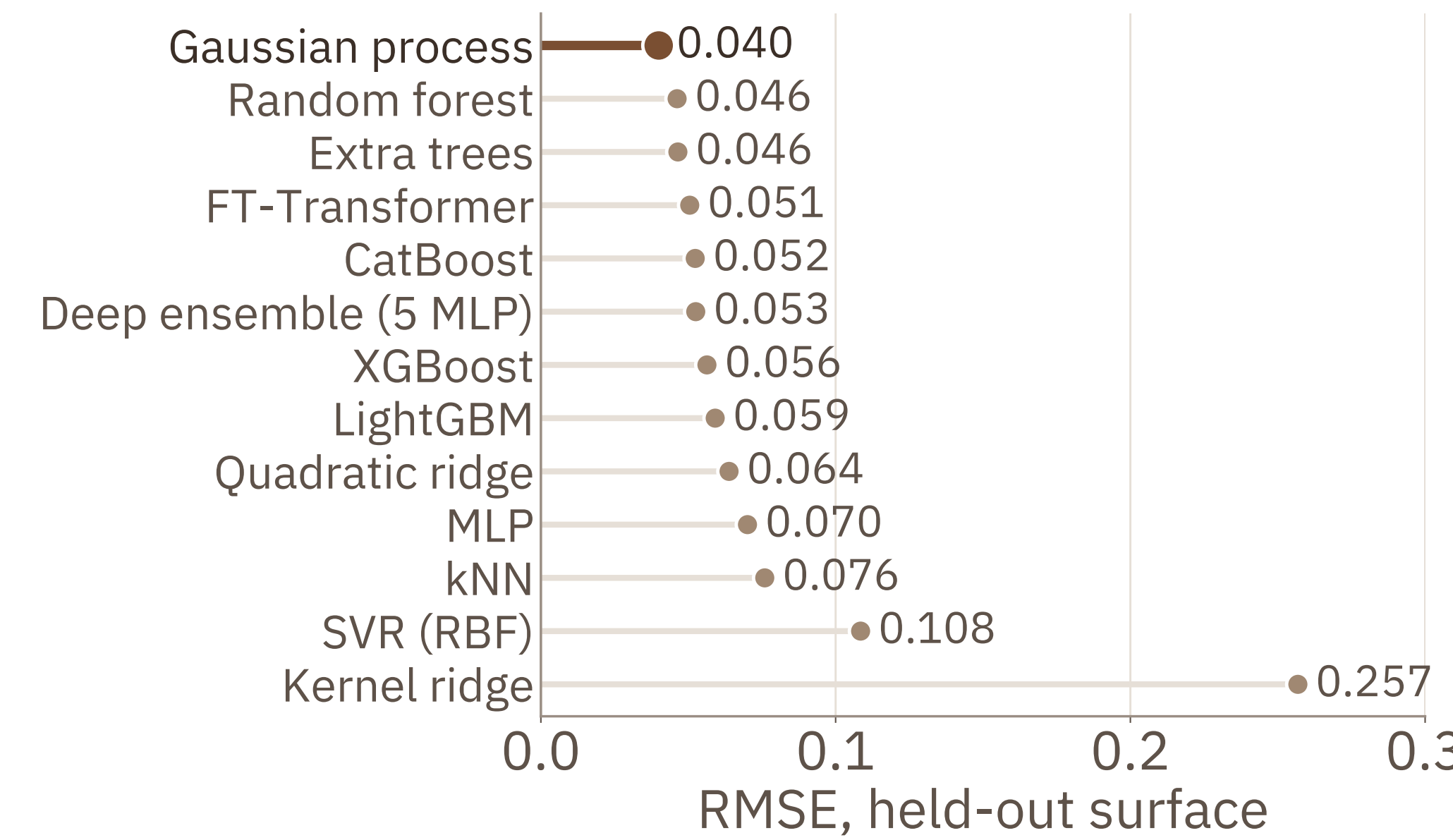
Each condition was repeated five times, so a random split puts near-identical twins in train and test.

- Grouped 5-fold CV** over 297 replicate groups: accuracy on seen surfaces.
- Leave-one-surface-out**, 12 folds: the model-selection criterion.
- REF-H held out**: never used to train or choose a model.

Inputs $\ln Re$, $\ln We$, $\ln D_0$, φ , V_{tex} ; target $\ln \beta_{\max}$. A change is accepted only if its 95% interval from **resampling whole surfaces** (4,000 draws) excludes zero.

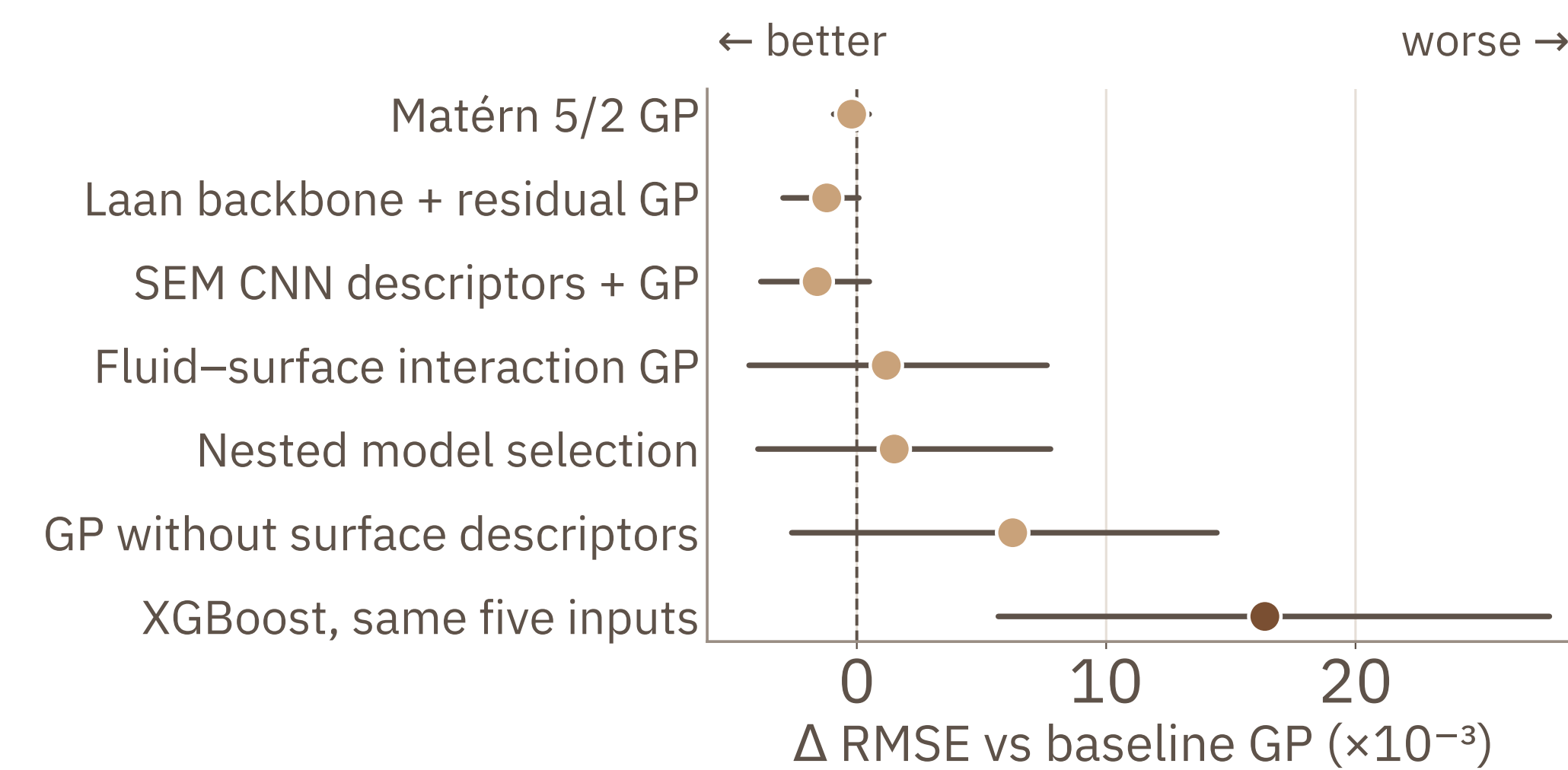
RESULTS

6 13 models, one test



0.040 Gaussian process (Matérn 3/2) on the held-out surface, 1.1% mean error, against 0.056 for XGBoost (95% CI of difference 0.006–0.028). On seen surfaces: 0.036.

7 What did not help

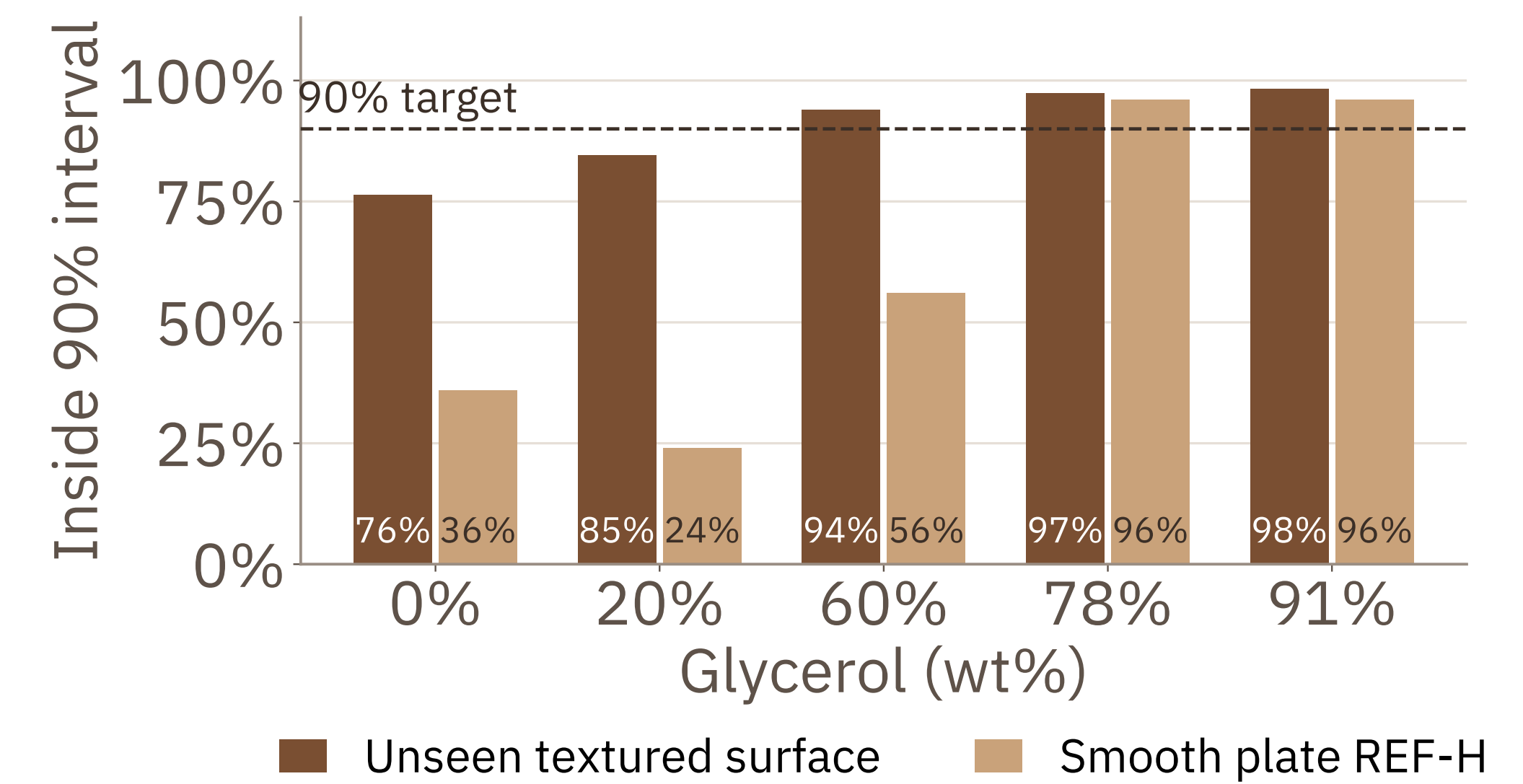


Paired surface-bootstrap 95% CIs. A Laan backbone, other kernels, nested selection and CNN descriptors each move the error by under 0.002. Only XGBoost differs significantly, and it is worse. Removing φ and V_{tex} : 0.040 $\rightarrow$ 0.046. **The simple GP stays.**

FUTURE WORK

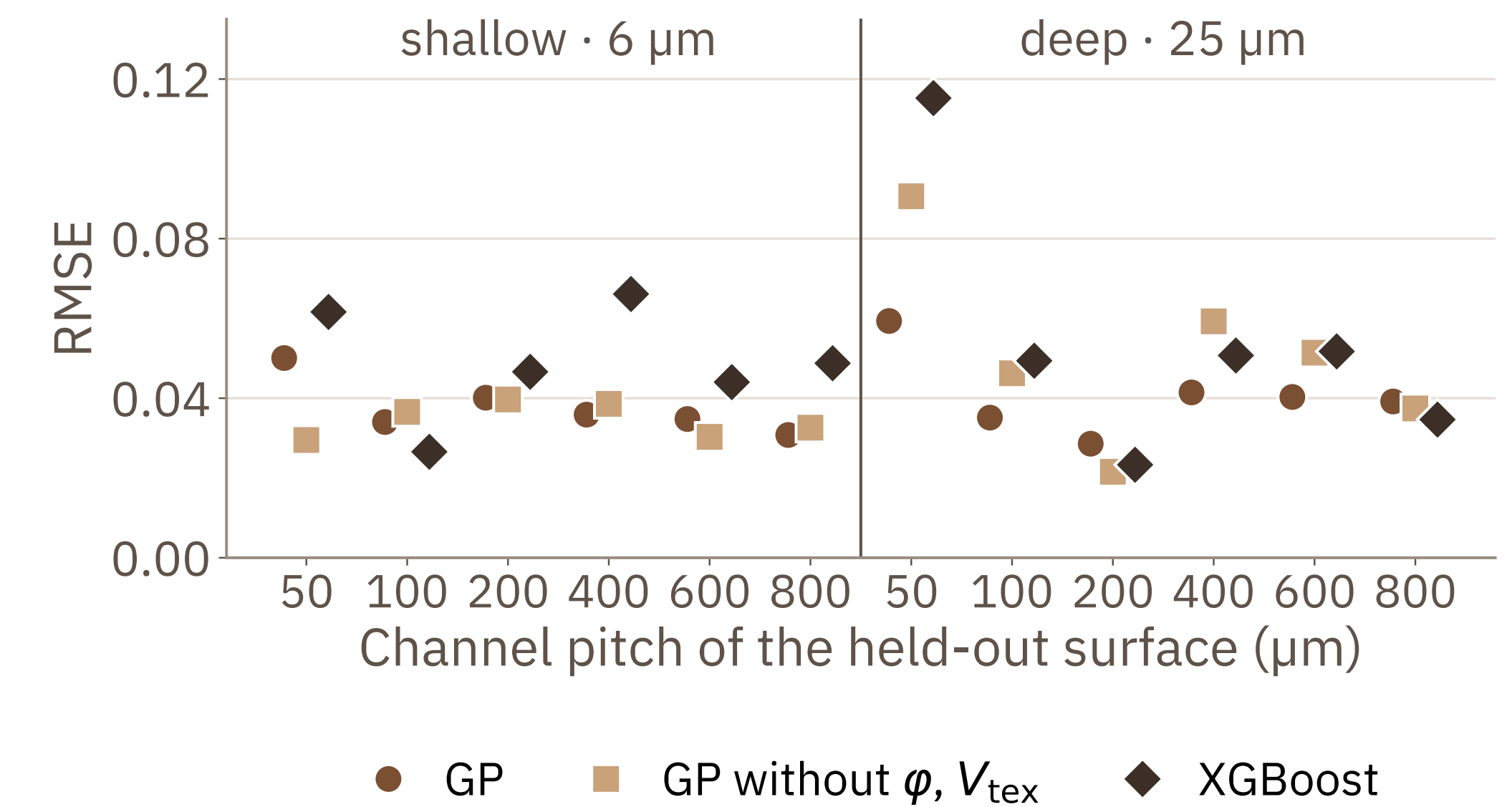
- Measured advancing and receding angles for every fluid–surface pair as an input.
- Noise that depends on viscosity (Ohnesorge number).
- New surfaces, with a test set held back before any tuning.

8 Knowing when not to trust it



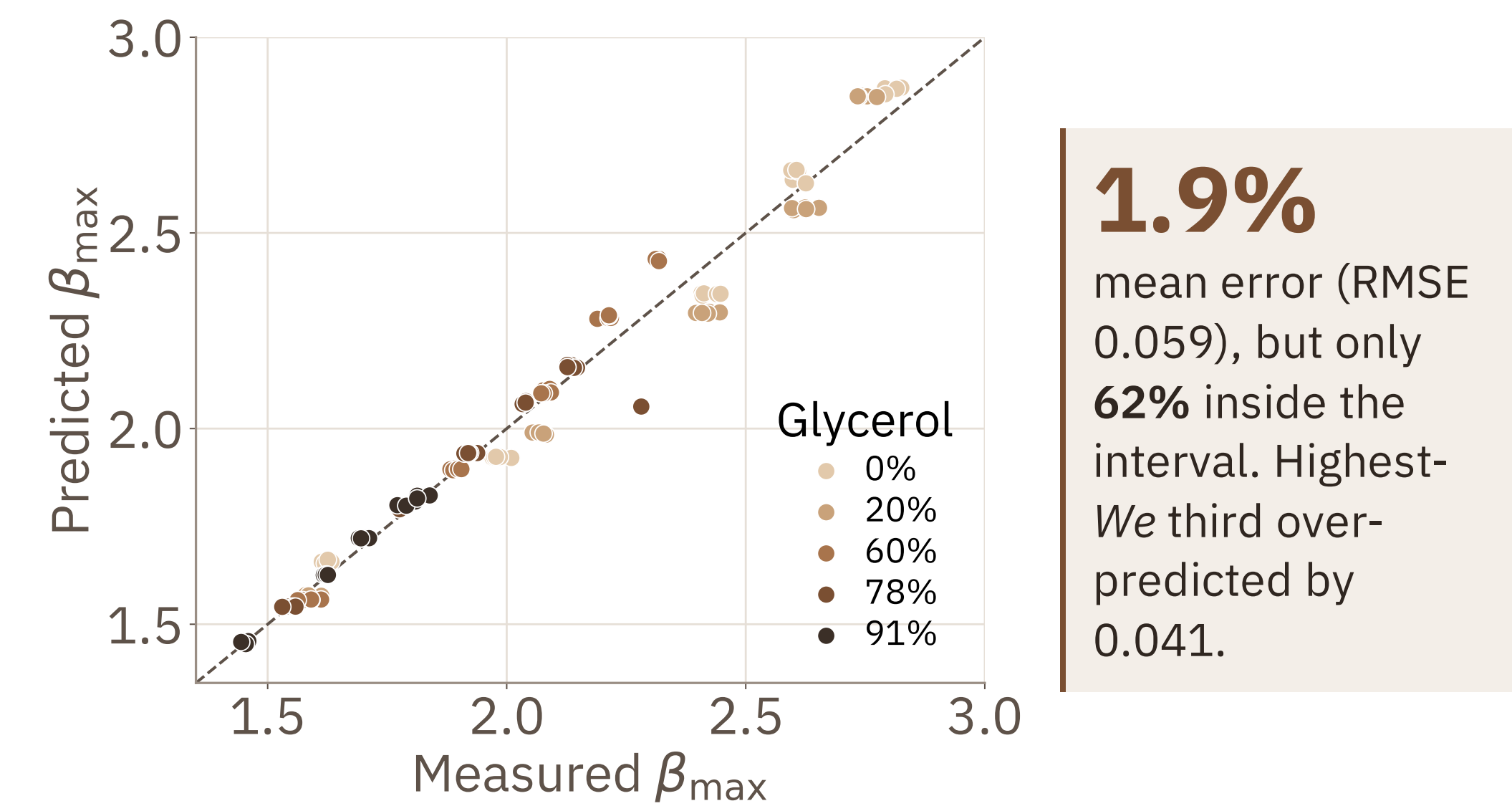
90.1% of unseen-surface impacts fall inside the 90% interval (84–96% per surface), but 76% for water against 98% for 91 wt% glycerol. The OOD check flags D50 and S50.

9 Surface by surface



D50 ($\varphi = 0.01$, the edge of the range) is hardest for every learner. The descriptors cut its error from 0.091 to 0.059, but hurt on S50 and D200. With 12 surfaces, a hint, not proof.

10 Smooth-plate stress test



Misses sit in water, 20% and 60% glycerol (24–56% coverage); viscous fluids stay at 96%. REF-H is hydrophobic, the textures superhydrophobic: **wettability is the missing input**. Less friction over trapped air may explain why textures spread further **HYPOTHESIS**

REFERENCES

- [1] M. Može et al., Data in Brief 61 (2025) 111697; doi:10.17632/wsh8rxwd38.1
- [2] S. Jereb et al., Biomimetics 10 (2025) 357
- [3] N. Laan et al., Phys. Rev. Applied 2 (2014) 044018